



Determination of emergency assembly point for industrial accidents with AHP analysis

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ABSTRACT

Major industrial accidents, which are a type of technological disaster, are very important due to the security risks and financial damages that threaten the environment and human health in today's industrialization. In this study, it was aimed to propose an approach that will guide the decision makers to choose the emergency assembly point that should be in the distance or shelter where the employees will be not affected by the negative consequences of emergencies within the scope of the obligation of industrial establishments preparing an internal emergency plan for major industrial accidents. For this purpose, in the first stage, modelling studies were carried out with ALOHA (Areal Locations of Hazardous Atmospheres) Software over possible accident scenarios in an industrial establishment containing different types and amounts of hazardous chemicals. As a result of modelling studies, possible toxic emissions, fire and explosion effect distances and threat zones for the industrial establishment were obtained. In the second stage, the weights of the main and sub-selection criteria to be used in determining the assembly point were calculated. This stage was carried out based on the comparison data obtained as a result of the questionnaire applied to professionals with the help of AHP (Analytic Hierarchy Process) method, which is one of the multi-criteria decision making methods. In the last stage, three candidate points were selected considering the physical effect areas determined in the first stage in the boundaries of the establishment, where the employees were evaluated to be affected the minimum from the negative consequences of industrial accidents. These candidate points were evaluated again with the AHP method on the basis of the sub-criteria whose relative weights were determined in the second stage and a selection was made. As a result, an approach that provides the solution of our problem was obtained.

1. Introduction

In recent years, irregular urbanization, industrialization, events such as unplanned industrialization and climate change have increased the number and effects of disasters. Disasters are examined under two main titles: natural disasters (slow developing-severe cold, drought and sudden developing-earthquake, flood, mass movement, landslide) and technological (human sourced) disasters (Isik et al., 2012). While natural disasters are disasters based on natural events, technological disasters are defined as global events and events caused by erroneous or incomplete planning and practices. Industrial accidents are evaluated under technological disasters. Many major industrial accidents due to the production, storage and use of hazardous chemicals were occurred in the historical process, and huge material and lethal losses that cannot be compensated after these (Feyzin 1966, Flixborough, 1974; Seveso 1976;

Bhopal 1984; Baia Mare, 2000; Turgut et al., 2011; Buncefield, 2005). Between 2000 and 2020, 5153 technological disasters took place in the world, 778 of which were in the industrial accident category, and 22 806 of the total 160 487 loss of lives occurred as a result of industrial accident (EM-DAT, 2020). Emergency plans and determination of the emergency assembly point are extremely important in the prevention of industrial accidents (Seveso III Directory, 2012). It is also an important issue to evaluate the consequences of explosion, fire, toxic emission of industrial accidents and to determine their physical effect distances. Consequences analysis is a part of the risk assessment and is a concept that should be considered primarily in the preparation of emergency plans. Environmental consequences analysis of chlorine release in a manufacturing facility in Nigeria was conducted by Sonibare and Ade-niran (2019). The analyzes were carried out with ALOHA Software, the effect distances in yellow, orange and red threat zones were determined

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as 2000–2500 m (2 kW/m²-pain within 60 s), 966–1300 m (5kW/m²-second degree burns within 60 s) and 289–382 m (10kW/m²-potentially lethal within 60 s), respectively. 457 tons of butane stored tank was analyzed with the PHAST modelling program by Nabhani et al. (2019), and it was stated that the most likely consequence was a jet fire (858 m). Yang et al. (2019) showed that the toxic release effects of a propylene storage tank after rupture had lethal toxicity results up to 199 m. The interaction of changes in volume of the methane in the atmosphere and the explosion events in semi-closed pipes with other explosion parameters were investigated by Jiang et al. (2016). It was shown that there was a linear relationship between high pressure, temperature, burst rate and burst distances due to fuel volume. An exposure distance of 2 km in diameter was detected for a tanker storing chlorine through different wind speed scenarios by Brzozowska (2016). It was stated by Ebrahemzadih et al. (2015) that the fire could be controlled within 15 m from an oil tank of the petrochemical industry after modelling studies. A safety assessment of methanol at a refinery was made by Xing and Wang (2015), and a leak at a distance of 100 m in 60 s was determined to have a lethal effect. The effect distances for ammonia were analyzed by ALOHA Software by Inanloo and Tansel (2015). Wind direction and atmospheric stability of alternative routes related to evacuation were evaluated and it was stated that the effect distances of the transportation made at night hours with routing and choosing the appropriate shipping amount had shorter effect distances compared to daytime hours. The consequences of an industrial accident were modeled with low oxygen concentration in the external environment by Kotek et al. (2014). In the scenario where liquid nitrogen was poured and then evaporated, the population was found to be more affected. The explosion accidents in benzene tanks were examined by Zhao and Tian (2013), and it was stated that the use of hazard modelling software had beneficial results in creating internal emergency plans. The environmental risks and threats caused by benzene leakage were investigated, and death and injury areas were determined by considering the benzene threshold values by Shao et al. (2013). In the study put forward by Levy and Taji (2007), condensation change, combustion and explosion area models were carried out after the release of ammonia stored in liquid phase. The changes on the impact areas were compared with the parameter changes made on the package software used to determine the effect distances. The consequences analysis of the tank containing liquid/gas propane mixture were studied by Xing and Wang (2012) and the BLEVE (Boiling Liquid Expanding Vapour Explosion) effect was determined at a distance of 2.2 km. The explosion of oil stored in liquid form was examined with ALOHA Software and its results were used in the quantitative risk assessment by Shuai et al. (2009). A hydrogen tank stored in liquid phase was analyzed by Rigas and Sklavounos (2005) and its effect areas were calculated up to 60 s with the “Event Tree Analysis Method”. The decision support system was used to simulate the consequences of industrial accidents, and vapour cloud explosion, pool fire and toxic emission effect distances were determined for a particular industrial facility by Quaranta et al. (2002). The BLEVE event was modeled on two closed vessels containing chemicals, and analyzes on thermal radiation and smoke propagation were made by Planas et al. (1998).

The problem of determining an assembly point among candidate points is a place selection problem. In the literature, especially in the field of disaster logistics, it has been determined that there are many studies for determining the location of the distribution center, disaster station and shelters. The affected area after natural or man-made disaster was divided into two categories as the location of the center where the victims of the disaster would be accommodated and humanitarian aid would be transported to them during this time by Sarma et al. (2018). Unmet demands in disaster management operation were addressed as cost items and their minimization problem was solved with the “LINGO 13.0” package program. Within the scope of short-term preparation for a flood disaster envisaged by Mejia-Argueta et al. (2018), a multi-criteria optimization model integrated with GIS was

established for the location of emergency facilities (distribution centers, shelters and assembly points), location, evacuation and distribution of humanitarian aid. The problem of locating disaster logistics warehouses, which ensure that emergency aid supplies were delivered to the points of need in the shortest time possible was addresses by Ayvaz and Kucukasci (2017). In the first stage of the study, a cluster coverage model determining the number of minimum number of alternative locations for a given coverage area, and in the second stage, the p-median problem for demand-weighted distance minimization were developed. The locations of facilities such as pre-and post-disaster situations, distribution centers, warehouses, shelters, garbage collection areas and health centers were determined to obtain the emergency humanitarian logistics facility location by Boonmee et al. (2017). The selection of storage location in disaster/humanitarian logistics was determined with the fuzzy AHP approach by Bolturk et al. (2016). The study was concluded with a case study in which one of the five potential regions that could be used as a storage area in the Marmara region was selected. The criteria that can be used in the selection of the distribution center location was weighted by the AHS method and the selection of the most suitable establishment location among the candidate points with the VIKOR method were carried out for Erzincan province by Peker et al. (2016). In the province of Duzce, using the fuzzy TOPSIS method, the most appropriate establishment location among the four candidate disaster stations was determined by Aslan et al. (2015). The problem of determining the number and location of humanitarian aid distribution centers to be established in the relevant region after the disaster was addresses by Abounacer et al. (2014) as a multi-purpose decision making problem. A solution proposal that covered the entire area affected by the disaster was proposed by minimizing the distribution center, vehicle and transportation times. A study on what qualitative and quantitative criteria will be used in the location of temporary shelters to be determined and how to establish the hierarchical structure of these criteria were put forward by Nappi and Souza (2014). The problem of determining disaster stations and assembly center among alternatives was addressed by Cicekdagi and Kiris (2012), and appropriate disaster station locations were determined using university campus area cluster analysis. A disaster logistics center site selection decision support system based on analytical hierarchy/fuzzy analytical hierarchy process methods was proposed by Turgut et al. (2011). Zhu et al. (2010) focused on the location and capacities of resource stores that can be used in emergencies, and a model to reduce the total cost was introduced. Erden and Coskun (2010) determined the criteria that can be used in the determination of the new locations of the fire stations, the Analytical Hierarchy Method (AHM) method was used, and the most appropriate location analysis was performed in the GIS media. The problem of determining the facility locations of medical supplies in relation to large-scale emergencies was formulated as a maximum coverage problem and three heuristic algorithms were used by Jia et al. (2007). The cluster coverage settlement model was applied by Hale and Moberg (2005) in choosing a safe facility location.

It was determined that there were a limited number of resources addressing issues such as evacuation of the community, determination of route information and routes for evacuation, location of shelter and location of the assembly point during an industrial accident. The options to implement an evacuation plan based on the emergency risk assessment or to apply shelter in the shelter inside or outside the place were evaluated following the toxic chemical (chlorine) leakage scenario by Yoo and Choi (2019). It was stated that it was safer to evacuate in 10.2 min for the local administration building (1390 m) and in 13.2 min for train station areas (1470 m) using correlations and GIS after chlorine release. An analysis study was carried out by taking 30 tsunami evacuation maps as a sample, and a suggestion was made on how to create evacuation routes depending on potential impact areas by Girres et al. (2018). An integrated emergency response model was developed for toxic gas evacuation accidents by Cao et al. (2016). The study was conducted for ammonia chemical and gas concentration distribution,

expected death and damage estimation and evacuation route information were shared. It was revealed that scenario based changes cause different maps to be produced in the tsunami hazard area, evacuation area and routes by Scheer et al. (2012). Guo et al. (2012) aimed to reduce the impact of nuclear accident and radiation leakage accidents on the population, and an evacuation planning was realized by developing an evacuation management model based on linear programming technique. Emergency response planning was carried out by the multi-purpose optimization model by Georgiadou et al. (2010). The options of evacuation, shelter protection, and continuation of normal activity were evaluated through case studies based on the impact areas created by BLEVE and toxic release events. Thermal radiation and BLEVE threat zones were determined with the worst case scenarios by using public exposure guides in order to develop an emergency response plan and determine a public protection zone by O'Mahony et al. (2008). A decision support system based on an analytical network process (ANP) was developed by Levy and Taji (2007) for the purpose of a chemical-induced accident evacuation. A decision support system that could communicate interactively between different environmental software tools was proposed and assembly points, evacuation zones and routes for the relevant region were determined through different scenarios by Kiranoudis et al. (2002) to reduce the impact of large-scale industrial accidents on the population. Chakraborty (2001) used methods based on environmental modelling and GIS integration to identify exposure zones, with almost 80% of schools and enrolled students in the relevant region reported to be in potentially worst exposure zones. The impact areas of hazardous substance release scenarios in order to evaluate the emergency response capabilities in relation to the risk areas located on the Mexico/USA border were determined by Cannon et al. (1998). In this context, civil evacuation routes, straits and security problems faced to create the related routes were presented. Accident scenarios for hazardous sources in the densely populated Ravenna region in Italy were developed by Egidi et al. (1995), and the total impact area was determined by the consequences analysis, threat zones and safe zones were defined for both the port environment and human dense areas.

The aim of this study is to determine the emergency assembly point (s) within the scope of the internal emergency plan for major industrial accidents with a new approach based on scientific facts that will guide decision makers and be applied to other similar location selection problems. This selection of the assembly points problem was addressed with the method of multiple-criteria decision analysis. The solution of the problem was initiated by determining the physical impact areas of potential industrial accidents that may occur due to hazardous chemicals in an exemplary industrial establishment related to explosion, fire and toxic emission. Then, the main and sub-selection criteria used to determine the emergency assembly point were weighted by the Analytical Hierarchy Process (AHP) analysis. Finally, an assembly point was chosen among candidate points in the sample establishment.

2. Materials and methods

The problem of selecting the assembly point for the internal emergency plan of industrial accidents with the multi-criteria decision-making method was carried out in three stages:

Stage 1: Determination of explosion, fire and toxic emission physical impact areas of hazardous chemical originated industrial accidents in an exemplary industrial establishment over hypothetical scenarios with ALOHA Software, Stage 2: Weighting the relevant selection criteria for the determination of the assembly point within the scope of the internal emergency plan with AHP analysis, Stage 3: The selection of the assembly point among the candidate points in the relevant industrial establishment again with AHP analysis.

2.1. Physical effects modelling studies

ALOHA (Areal Location of Hazardous Atmospheres) Software which is provided by the US Environmental Protection Agency and is intended for utilize by users for pre-planning and in emergency response. ALOHA has its own extensive library that covers over 700 chemicals (pure substances and a small number of mixtures/compounds). This program produces scenarios based on location, chemical, atmospheric conditions and choice of chemical source type (direct, puddle, tank, gas pipe) and provides hazard source centered threat zones and distances. It is easy to use and provides a rapid output. The ALOHA Software enables the detection of the effect distance up to 10 km. The grid image of impact area which is obtained as a model output, can be transferred to MARPLOT or GIS easily. The red threat zone represents the worst hazard level, orange and yellow respectively representing decreasing hazard levels. For this reasons, ALOHA Software which is freely available, was used in modelling studies. Three selections must be defined for modelling in ALOHA Software:

2.2. Location and chemical selection

Modelling studies were carried out for an industrial establishment containing butane, propane, LPG (Liquefied Natural Gas), MTBE (Methyl tert-butyl ether) hazardous chemicals. It was assumed that these chemicals were located in different parts of the same establishment.

2.3. Atmospheric conditions

The average atmospheric conditions of the location of the industrial establishment required for the modelling study are given in Table 1 according to the annual observation records.

2.4. Chemical source and scenario selection

Modelling studies were carried out over three different hypothetical scenarios for tanks of different volumes including butane, propane, LPG, MTBE chemicals.

Scenario (1): Leaking tank, chemical is not burning as it escapes into the atmosphere.

Scenario (2): Leaking tank, chemical is burning as a jet fire.

Scenario (3): BLEVE (Boiling Liquid Expanding Vapour Explosion), tank explodes and chemical burns in a fireball.

2.5. AHP analysis

7 main criteria, each of which was divided into sub-criteria, were determined for the decision problem, the general purpose of which was to determine the assembly point. The decision hierarchy of the emergency assembly point selection and sub criteria for emergency assembly point evaluation are given in Fig. 1 and Fig. 2, respectively.

The foundations of AHP were laid in 1968 by Myers and Alpert. Saaty described this method as a hierarchical structure of the components of a decision problem and turned it into a systematic process sequence (Mert et al., 2010). The steps performed while applying AHP method were

Table 1
Average atmospheric conditions.

Air temperature	18.1 °C
Wind speed	2.7 m/s
Cloud cover	Clear
Ground roughness	Urban or forest
Relative humidity	75%
Wind is from	NW
Atmospheric stability class	D
Measurements height above ground	2 m
Inversion	N/A

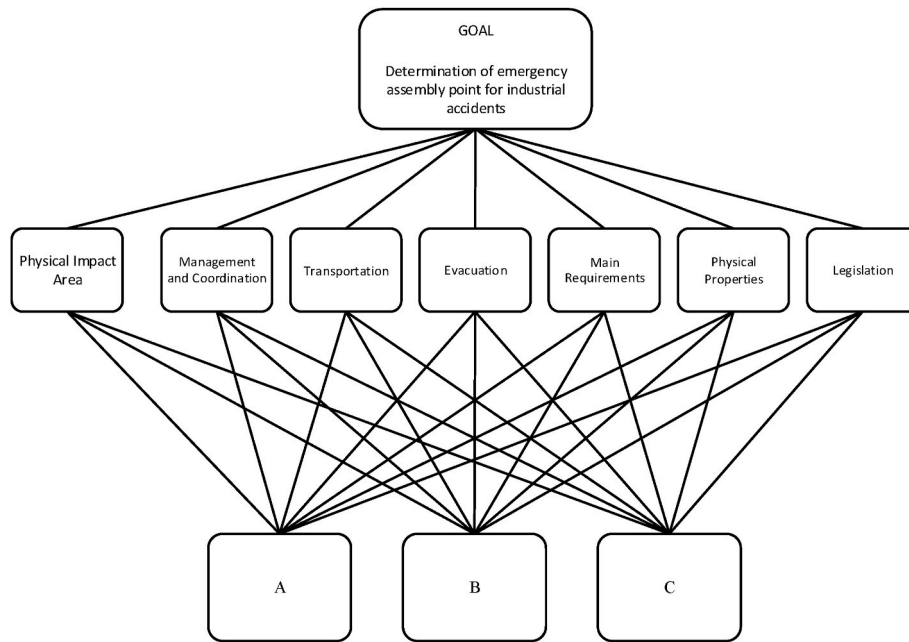


Fig. 1. The decision hierarchy of the emergency assembly point selection.

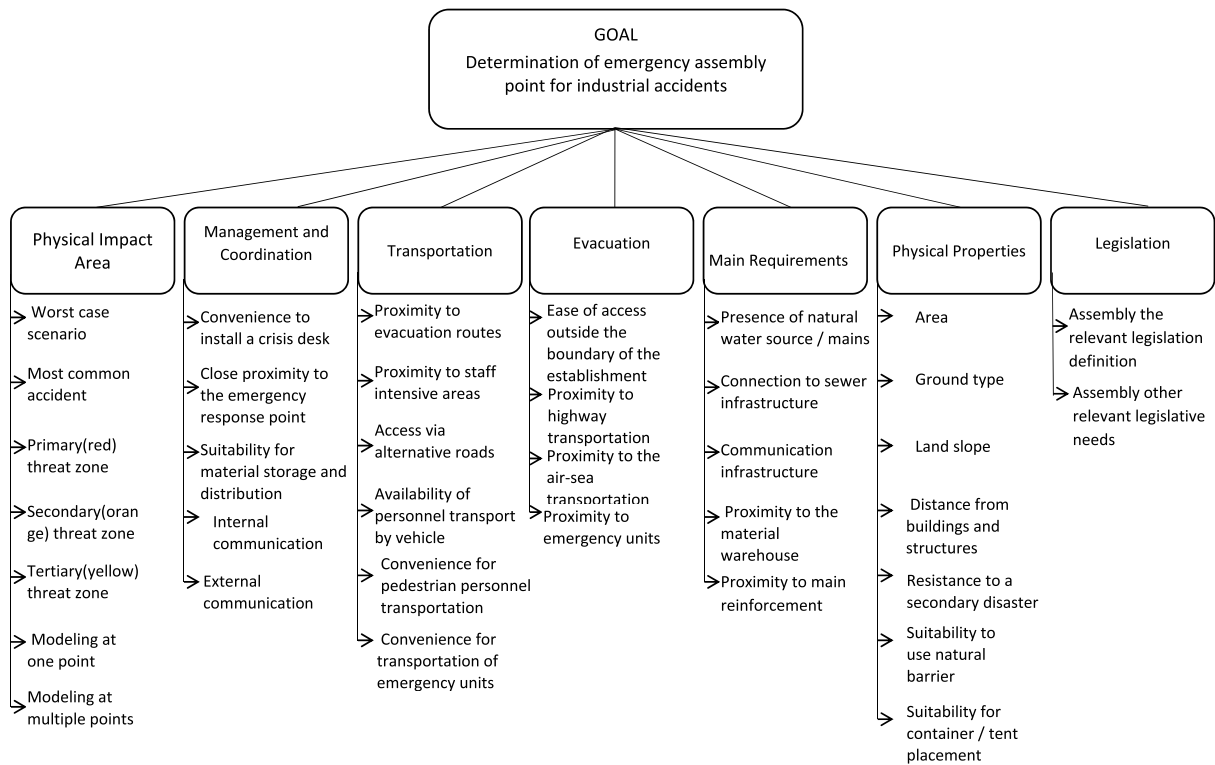


Fig. 2. Sub criteria for Emergency Assembly Point Evaluation.

explained below with its formulation.

Step 1: Identify the decision making problem.

In the first stage, the goal that the decision-maker wants to reach was determined. With this purpose, the criteria and sub-criteria that will be ranked according to their relative importance were determined and hierarchical structure was created by introducing alternatives.

Step 2: Creating cross-benchmark matrices.

The matrices created show the relative superiority of the criteria and sub-criteria against each other (Eq. (1)). The importance levels of the

criteria were filled according to the Saaty importance scale in Table 2 and the values on the matrix diagonal were assigned as “1”.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad (1)$$

Table 2

The scale of importance of Saaty.

Quantitative value	Qualitative variables	Explanation
1	Equal Important	Two factors contribute equally to the goal.
3	One is less important than another	Experience and judgment make one factor less preferable to another.
5	Strongly important	Experience and judgment strongly favour one factor over another.
7	Very strongly important	A factor is strongly preferred and its predominance is comfortable in practice.
9	Extremely important	Evidence for the preference of one factor over another has tremendous reliability.
2,4,6,8	Intermediate values	Preference values are the values used when close/need to compromise.

Step3: Calculation of the eigenvector showing the superiority of the criteria relative to each other.

The comparison matrix created shows the importance of the criteria according to each other. The eigenvector was used to determine the percentage significance distributions within these criteria. This vector was calculated as in Eq. 2 and B column vector was obtained.

$$b_{ij} = \frac{a_{ij}}{\sum_{i=1}^n a_{ij}} \quad (2)$$

When the relevant stage was repeated within all evaluation criteria, B column vectors were obtained as much as the number of criteria. When “n” B column vectors were put together, C matrix was created as shown in Eq. (3).

$$C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{bmatrix} \quad (3)$$

Percentage significance distributions of the criteria were determined by using the created C matrix. The vector of W was calculated by taking the arithmetic average of the line components that make up the C matrix, as shown in Eq. (4).

$$w_i = \frac{\sum_{j=1}^n c_{ij}}{n} \quad (4)$$

Step 4: Calculation of the consistency rates (CR) of the benchmark matrices.

In AHP method, CR value should be maximum 0.10. In this way, matrices with a consistency ratio of 0.10 or less are considered consistent. Other matrices are considered inconsistent, indicating that the decision-maker's inconsistency or calculation error has been made. The essence of this calculation was based on comparing the number of criteria and a coefficient called “Basic Value” (λ). To calculate λ , firstly the column D was obtained as shown in Eq. (5) from the matrix multiplication of the comparison matrix A and the column vector W.

$$D = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & w_1 \\ a_{21} & a_{22} & \dots & a_{2n} & w_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & w_n \end{bmatrix} \quad (5)$$

The basic value (E) for each criterion was obtained from the division of the reciprocal elements of the column D vector and column W vector (Eq. (6)). The arithmetic mean of these values was calculated as shown in Equation (7) and given λ for comparison.

$$E_i = \frac{d_i}{w_i} \quad (i = 1, 2, \dots, n) \quad (6)$$

$$\lambda = \frac{\sum_{i=1}^n E_i}{n} \quad (7)$$

After λ was calculated, the “Consistency Indicator (CI)” was calculated using Equation (8).

$$CI = \frac{\lambda - n}{n - 1} \quad (8)$$

In the next step, CI was divided by the standard correction value called “Randomness Indicator (RI)” and given in Table 3, “Consistency Rate (CR)” was obtained (Eq. (9)).

$$CR = \frac{CI}{RI} \quad (9)$$

Step 5: Calculation of the result vector at the decision point.

At this stage, percentage significance distributions of decision points were calculated for each criterion. After each comparison process, S column vectors were obtained showing the percentage distributions of each criterion according to the decision points. The K decision matrix shown in Equation (10), consisting of “n” S column vectors, was created.

$$K = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ s_{21} & s_{22} & \dots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{m1} & s_{m2} & \dots & s_{mn} \end{bmatrix} \quad (10)$$

When the resulting decision matrix was multiplied by the W column vector, the L column vector shown in Equation (11) was obtained. This column vector represents the percentage distribution of decision points and also shows the order of importance of decision points.

$$L = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} & w_1 & l_{11} \\ s_{21} & s_{22} & \dots & s_{2n} & w_2 & l_{21} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ s_{m1} & s_{m2} & \dots & s_{mn} & w_n & l_{m1} \end{bmatrix} \quad (11)$$

As a requirement of the AHP application, comparison matrices were created to find the weights of the main criteria and sub-criteria. Comparison data was obtained as a result of the questionnaire applied to professionals who dominating national/international legislation on industrial accidents and their prevention or mitigation. Relevant professionals consisted of 3 public personnel, 2 academicians and 1 industrial organization employee. Since the opinions of more than one decision maker were taken into consideration, the study can be evaluated under the heading of “group decision making”.

2.6. Determination of emergency assembly point for industrial accidents

The assembly point was determined by multiplying the importance of the alternative on the basis of the sub-criterion, the relative weight of the sub-criterion and the relative weight of the main criterion (Eq. (12)).

$$S_i = \sum_{j=1}^n \quad (12)$$

S_i : Relative weight indicating the ranking of the 1st alternative as a result of AHP

w_i : The weight of the main criterion i

v_{ij} : The weight of the jth sub criterion of the ith main criteria

Table 3

Randomness indicator values.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0.00	0.58	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.54	1.56	1.57	1.59

p_{ji} : Weight of the j th sub criterion for the 1st main criterion for the i th alternative

$i = 1, 2 \dots n, j = 1, 2 \dots m$.

The choice of the assembly point for the internal emergency plan of industrial accidents was realized by evaluating the 3 candidate points determined in different parts of the industrial establishment with again the AHP method and creating the binary comparison matrices of the sub-criteria.

3. Results

3.1. Physical effects modelling studies

In modelling studies, it was assumed that the chemicals were located in different parts of the same establishment and showed all the effects (explosion, fire, toxic spread) after a possible industrial accident. Modelling studies were carried out for an industrial establishment that stored butane, propane, LPG and MTBE chemical tanks. These modelling studies were carried out on three different scenarios with the ALOHA Software.

3.1.1. Propane tank

Propane gas safety sign with NFPA (National Fire Protection Association) rating of 2 for health, 4 for flammability and 0 for instability. For propane, it is inevitable that fire and explosion occur in the presence of an ignition source and sufficient oxygen (Kao and Hu, 2002).

3.1.2. Butane tank

Butane gas safety sign with NFPA rating of 1 for health, 4 for flammability and 0 for instability. As it is a gas that is heavier than air, it may accumulate in low areas, cause oxygen deficiency, move on the ground and ignite at a point away from its source.

3.1.3. LPG tank

LPG safety sign with NFPA rating of 1 for health, 4 for flammability and 0 for instability. LPG is flammable in a wide concentration range and may cause various hazards due to its physical properties. LPG gas is heavier than the air mixture and a flame can reach LPG leak source very easily. It is colourless and odourless, making leak detection difficult. Storage equipment can be severely ruptured and rocketed if exposed to heat for a long time. The movement of LPG in the air is directly related to the wind and ground roughness. While a strong wind can dissipate LPG vapour in a short time, light wind does not dissipate the vapour that is heavier than air, but it can drag at ground level and cause it to accumulate in places such as pits/ditches on the ground. ALOHA Software does not allow calculation for special chemical mixtures. In this study, calculations were carried out with the higher concentration chemical of the LPG tank.

3.1.4. MTBE tank

MTBE safety sign with NFPA rating of 1 for health, 3 for flammability and 0 for instability. It is often used as an octane booster in gasoline. MTBE can contaminate groundwater or soil by leaking or pouring from underground and aboveground storage tanks. Low MTBE levels cause an unpleasant taste and odour in the water, making drinking water resources unusable. Higher MTBE levels are a potential carcinogen for humans.

Modelling studies were carried out on the related scenarios

(Scenarios 1–3) considering that the chemicals were stored in tanks -propane (53 200 kg) in a vertical cylindrical tank (diameter: 3 m, height: 15 m), butane (352 300 kg) in a vertical cylinder tank (diameter: 10.5 m, height: 7 m), LPG (2 900 000 kg) in a horizontal cylinder tank (diameter: 21.2 m, height: 14.1 m) and MTBE (22 300 000 kg) in a vertical cylinder tank (diameter: 45.72 m, height: 19.2 m) without a dyke area- and that the tanks were leaking. Information entries were made to the ALOHA Software regarding the type, amount of each hazardous source, the shape, size and location of the leak on the storage tank as part of the scenario.

After the scenario selection for the threat zone to be analyzed, “exposure limits” are defined. One or more toxic exposure limits can be selected when modelling a chemical release and evaluating the threat of toxicity in ALOHA Software. This exposure limit (Levels of Concern-LOC) represents the value of the relevant chemical exposure level (threshold concentration) when breathing for a certain period of time (exposure time) in a toxic release. In general, the lower the LOC value for a chemical, the more toxic the substance is. At this stage, ALOHA’s default LOC values can be used or the user can choose their own exposure limits. For each selected LOC, ALOHA estimates a threat zone, whose concentration is expected to exceed this LOC sometime after the release begins. ALOHA Software prefers one of the general public exposure guidelines values when determining the LOCs used as assumptions for toxic emissions. These guidelines are specifically designed to predict how this event will have an impact on humans and the environment in case of a one-time release. The most common public exposure guidelines: AEGs (Acute Exposure Guideline Levels) developed by the U.S. Environmental Protection Agency (EPA), ERPGs (Emergency Response Planning Guidelines) developed by the U.S. Emergency Response Planning Committee and the TEELs (Temporary Emergency Exposure Limit) developed by the US Emergency Management DOE Office. A toxic emission threat zone of a specified chemical may include one or all of these exposure values. If the AEGs and ERPG values cannot be displayed in the LOC options of the relevant chemistry, it indicates that these values are not included in the ALOHA library and in this case PAC (Protective Action Criteria for Chemicals-Protective Action Criteria) values are displayed. Each public exposure guide has three exposure values (layers) in itself, and although there are some fundamental differences between them, in general, the first layer (e.g. AEGs-1) represents a slight impact threshold, the second layer (e.g. AEGs-2) represents an escape-distortion threshold, and the third layer (e.g. AEGs-3) is a life-threatening threshold. The LEL (Lower Explosion Limit-Lower Explosion Limit) defined for the flammable zone refers to the lowest amount of flammable vapour in the air. High pressure effects for the vapour cloud explosion area in the ALOHA Software are evaluated in three stages as red threat zone-8.0 psi (destruction of buildings), orange threat zone-3.5 psi (serious injury likely) and yellow threat zone-1.0 psi (shatters glass). Red threat zone: 10 kW/m²-potentially lethal within 60 s, orange threat zone: 5 kW/m²-second degree burns within 60 s, yellow threat zone: 2 kW/m²-pain within 60 s correspond to thermal radiation in ALOHA (Cetinyokus, 2017).

As a result of modelling studies, the effect distance values of the relevant tanks are presented in Table 4.

It was seen from Table 4 that the effect distances varied depending on the type and amount of chemicals stored in the tanks and also the selection of scenario. The largest effect distances (1.80 km–3.90 km) were obtained at the end of Scenario-3 in the MTBE tank. The minimum effect distance was determined as 114 m in the red threat zone, representing blast area of vapour cloud explosion-overpressure effect distance, at the end of Scenario- 1 for the same tank. In the tanks containing propane,

Table 4
Effect distance values determined for different scenarios of the chemical tanks.

Chemical Tank	Scenarios	Red Threat Zone	Orange Threat Zone	Yellow Threat Zone
Propane	Scenario (1): Leaking tank, chemical is not burning as it escapes into the atmosphere			
	Toxic area of vapour cloud (Toxic effect distance)	381 m	538 m	924 m
	Flammable area of vapour cloud (Flammable effect distance)	624 m	–	1.40 km
	Blast area of vapour cloud explosion (Overpressure effect distance)	571 m	709 m	1.40 km
	Scenario (2): Leaking tank, chemical is burning as a jet fire			
	Thermal radiation effect distances	181 m	265 m	417 m
	Scenario (3): BLEVE (Boiling Liquid Expanding Vapour Explosion), tank explodes and chemical burns in a fireball			
	Thermal radiation effect distances	486 m	685 m	1.10 km
	Scenario (1): Leaking tank, chemical is not burning as it escapes into the atmosphere			
	Toxic area of vapour cloud (Toxic effect distance)	173 m	296 m	524 m
Butane	Flammable area of vapour cloud (Flammable effect distance)	393 m	–	1 km
	Blast area of vapour cloud explosion (Overpressure effect distance)	363 m	508 m	1 km
	Scenario (2): Leaking tank, chemical is burning as a jet fire			
	Thermal radiation effect distances	121 m	193 m	315 m
	Scenario (3): BLEVE (Boiling Liquid Expanding Vapour Explosion), tank explodes and chemical burns in a fireball			
	Thermal radiation effect distances	878 m	1.20 km	1.90 km
	Scenario (1): Leaking tank, chemical is not burning as it escapes into the atmosphere			
	Toxic area of vapour cloud (Toxic effect distance)	200 m	342 m	605 m
	Flammable area of vapour cloud (Flammable effect distance)	454 m	–	1.20 km
	Blast area of vapour cloud explosion (Overpressure effect distance)	415 m	578 m	1.20 km
LPG	Scenario (2): Leaking tank, chemical is burning as a jet fire			
	Thermal radiation effect distances	135 m	215 m	351 m
	Scenario (3): BLEVE (Boiling Liquid Expanding Vapour Explosion), tank explodes and chemical burns in a fireball			
	Thermal radiation effect distances	1.70 km	2.40 km	3.70 km
	Scenario (1): Leaking tank, chemical is not burning as it escapes into the atmosphere			
	Toxic area of vapour cloud (Toxic effect distance)	182 m	825 m	4.0 km
	Flammable area of vapour cloud (Flammable effect distance)	137 m	–	506 m
	Blast area of vapour cloud explosion (Overpressure effect distance)	114 m	156 m	340 m
	Scenario (2): Leaking tank, chemical is burning as a jet fire			
	Thermal radiation effect distances	172 m	241 m	371 m
MTBE	Scenario (3): BLEVE (Boiling Liquid Expanding Vapour Explosion), tank explodes and chemical burns in a fireball			
	Thermal radiation effect distances	1.80 km	2.50 km	3.90 km

butane and LPG the largest effects distances were determined as thermal radiation effects at the end of Scenario- 3. ALOHA is a Software that can be easily integrated with GIS and offers the user intuitive visualization and spatial analysis to evaluate the consequences of major industrial accidents. In this context, the threat zones and “Google Earth Pro” images of the modelling studies carried out over the Scenarios (1–3) for the same amount of MTBE tank as ALOHA Software are given in Fig. 3.

It was seen from Fig. 3 that the threat zones of the MTBE tank remained within the boundary of the establishment for all scenarios except Scenario 3.

3.2. AHP analysis

Binary comparisons for AHP implementation were discussed in two basic stages. In the first stage, the main criteria (Table 5) and in the second stage the sub criteria for each main criterion were compared among themselves (Table 6). Binary comparison matrices were determined by calculating the geometric averages, as they were obtained by the assessment of more than one person to combine judgments.

Higher weights of the main criteria of the emergency assembly point selection problem from Table 5 were determined as transportation, physical impact area and evacuation, respectively. Lower criterion weights were determined at the same rate for management and coordination and legislation. It can be said that the emergency assembly point in the facility was generally considered as a temporary assembly point by the experts, therefore it was thought that it should be an area where evacuation and transportation were effectively provided, away from the physical effects of the accident although it varied depending on the severity of the accident. Other criteria (Management and Coordination, Main Requirements, Physical Properties) may have been found less important in the selection problem by experts because assembly point was not considered as an area where long periods of time pass. Lack of sufficient explanation in the legislation on how to choose the assembly point and its properties may have caused the weight of this criterion to be determined as low by experts.

As can be seen from the tables, the “consistency rate” in the binary comparisons matrix was reached an acceptable level of <10% (0.100). As a result of the binary comparison of the main criteria for the specified purpose, it was determined that the most important criterion for determining the assembly point was the “transportation” criterion, which had 25% significance. As a result of the binary comparison of the sub-criteria related to the “physical impact area” criterion, which was the first of the main criteria, it was determined that the most important criterion was the “worst case scenario” criterion with 32% of significance. On the other hand, the most important criterion for the “management and coordination” criterion was the “external communication” criterion with 25% of significance, the most important criterion for the criterion of “transportation” was the “convenience of transportation of emergency units” with 27% of significance, and the most important criterion for the “evacuation” criterion was the “ease of access outside the border of the establishment” criterion with 44% of significance. In addition to these, the most important criterion regarding the “Main Requirements” criterion was the criterion of “natural water source/ mains water presence” with 37% of significance, the most important criterion regarding the “Physical properties” criterion was the criterion of “distance from buildings and structures” with 29% of significance and the most important criterion regarding the “Legislation” criterion was the “assembly the relevant legislation definition” criterion with 67% of significance.

3.3. Determination of emergency assembly point for industrial accidents

In order to determine the assembly point for the internal emergency plan of industrial accidents among candidate points in the sample establishment, 3 candidate points (A, B, C) were determined in different parts of the establishment in the distance or shelter where the employees

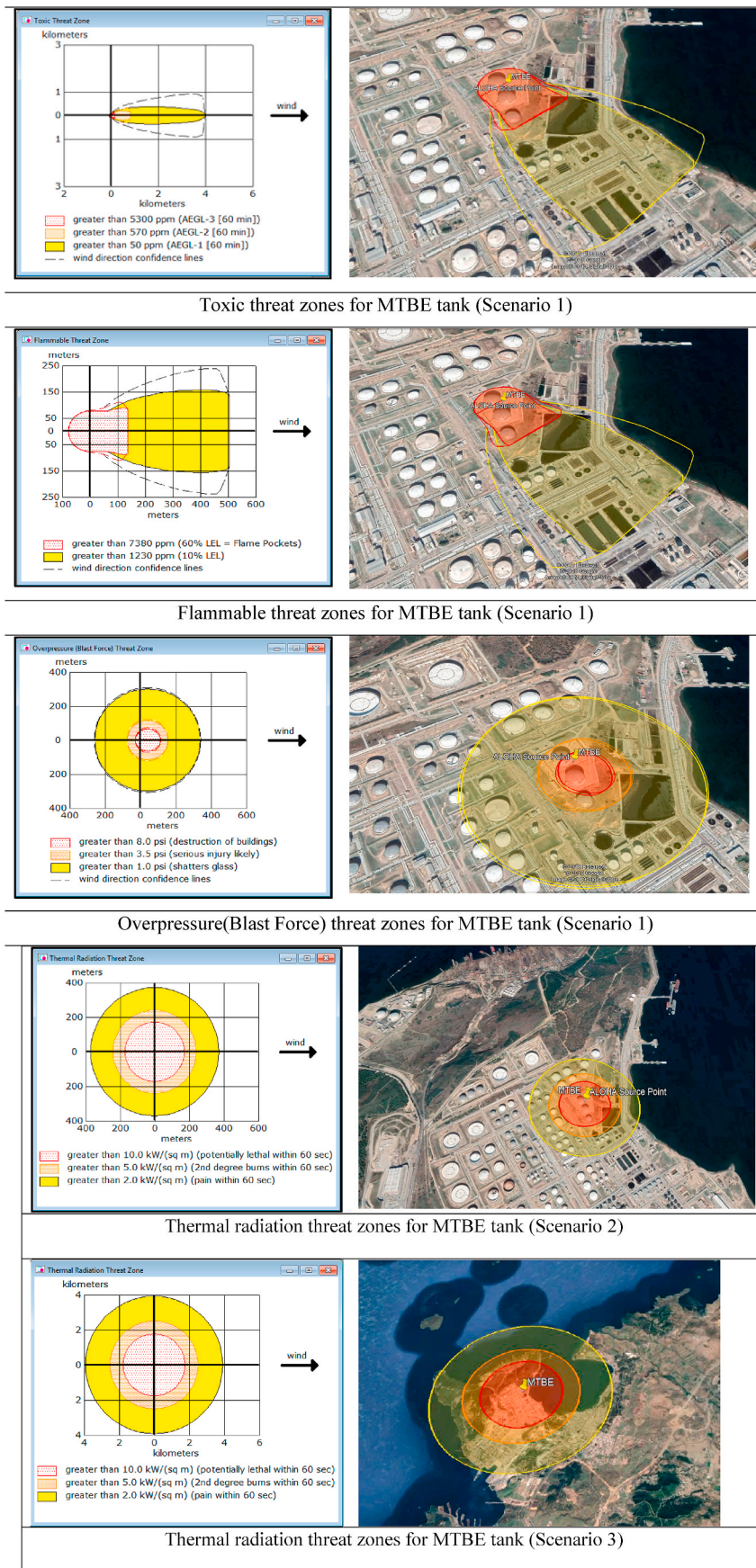


Fig. 3. Threat zones and “Google Earth Pro” images of the modelling studies carried out over the Scenarios (1–3) for the MTBE tank.

Table 5

Weight values of the main criteria.

Criteria	Weight Value/Priority Value
Main Criteria	
Physical Impact Area	0.24
CR = 0.028 < 0.1	
Management and Coordination	0.05
Transportation	0.25
Evacuation	0.21
Main Requirements	0.11
Physical Properties	0.09
Legislation	0.05

Table 6

Weight values of sub criteria.

Main Criteria	Sub Criteria	Weight Value/Priority Value
Physical Impact Area		
CR = 0.037 < 0.1	Worst case scenario	0.32
	Most common accident scenarios	0.23
	Primary (red) threat zone	0.17
	Secondary (orange) threat zone	0.08
	Tertiary (yellow) threat zone	0.05
	Modelling in one point	0.08
	Modelling at multiple points	0.07
	Convenience to install a crisis desk	0.19
Management and Coordination		
CR = 0.033 < 0.1	Close proximity to the emergency response point	0.17
	Suitability for material storage and distribution	0.17
	Internal communication	0.22
	External communication	0.25
Transportation		
CR = 0.021 < 0.1	Proximity to evacuation routes	0.24
	Proximity to staff intensive areas	0.10
	Access via alternative roads	0.17
	Convenience for pedestrian staff transportation	0.13
	Availability of personnel transport by vehicle	0.09
	Convenience for transportation of emergency units	0.27
Evacuation		
CR = 0.044 < 0.1	Ease of access outside the boundary of the establishment	0.44
	Proximity to highway transportation routes	0.20
	Proximity to the air-sea transportation route	0.09
	Proximity to emergency units	0.27
Main Requirements		
CR = 0.058 < 0.1	Presence of natural water source/ mains water	0.37
	Connection to sewer infrastructure	0.04
	Communication infrastructure	0.34
	Proximity to the material warehouse	0.18
	Proximity to main reinforcement areas	0.07
Physical Properties		
CR = 0.017 < 0.1	Area	0.09
	Ground type	0.08
	Land slope	0.06
	Distance from buildings and structures	0.29
	Resistance to a secondary disaster	0.21
	Suitability to use natural barrier	0.16
	Suitability for container/tent placement	0.11
Legislation		
CR = 0.000 < 0.1	Assembly the relevant legislation definition	0.67
	Assembly other relevant legislative needs	0.33

will not be affected by the negative consequences of emergencies. The image of the relevant candidate points and candidate points within the primary (red) threat zones determined as a result of physical impact modelling are given in Fig. 4a and Fig. 4b, respectively. The selection was made by creating binary comparison matrices of alternatives based

on sub-criteria, whose weights were determined as required by AHP application (Table 7).

The determination of the relative weights of each alternative for the purpose of determination of the emergency assembly point was calculated by multiplying the importance of the alternative on the basis of the sub-criterion, the relative weight of the sub-criterion and the relative weight of the main criterion. When the relevant calculations were carried out, the evaluation points obtained with the AHP method of the assembly point alternatives were determined as follows.

$$S_1 = 0.45 \text{ (Point A).}$$

$$S_2 = 0.36 \text{ (Point B).}$$

$$S_3 = 0.19 \text{ (Point C).}$$

As a result of the calculations the highest evaluation score was determined at point A (0.45), which was the parking area at the entrance of the establishment. This was followed by point B (0.36), which was the parking lot located near the cafeteria and social facilities in the second place, and lastly point C (0.19), which was the tanker filling porch of the facility.

4. Conclusion

Major industrial accidents, which is under the concept of technological disaster, have an important place in our lives due to the production of chemical substances, in which industrialization has accelerated and its use has become inevitable, especially in modern processes. A problem that may arise during the use, storage, transportation or disposal of these chemicals may cause fire, explosion and toxic emission that cause material damage, pose safety risks to human health and life, and require emergency response. Although such accident risks cannot be eliminated altogether, it is possible to minimize their effects with a good planning and precautions. In this study, an approach was proposed to guide decision makers in order to choose an assembly point that should be determined by the enterprises within the scope of the internal emergency plan for major industrial accidents, where the employees will not be affected by the negative consequences. In the first stage, modelling studies with different accident scenarios were carried out for hazardous chemicals stored in different types and amounts in an industrial establishment and physical impact areas of explosion, fire and toxic emission were determined with ALOHA Software. In the second stage, in order to weigh the main and sub-selection criteria used in determining the assembly point comparison data were obtained as a result of the questionnaire applied to professionals and relative importance weights were determined by AHP analysis. At the last stage, 3 candidate points determined within the boundaries of the establishment considering the physical impact areas at the distance that the employees will see the minimum impact from the negative consequences of emergencies, were evaluated using again the AHP method and a selection was made. As a result of the study.

- It has been observed that the method of storage, amount and type of hazardous chemicals directly affected the physical impact distances associated with toxic emission, fire and explosion that may occur after an industrial accident.
- AHP analysis, which is one of the MCDM (Multi-criteria Decision Making) methods defined as a sub-branch of decision analysis science, was used to determine the assembly point location for the internal emergency plans. It has been determined that AHP analysis is a suitable approach for determining the weight of selection criteria in such a selection problem, which aims to meet more than one criteria.
- The AHP method applied in the study was able to include both qualitative and quantitative evaluation criteria. Due to its structure based on binary comparison matrices of multiple decision makers, AHP has created the opportunity to evaluate the opinions and judgments represented by different experiences at the same time and in this sense, objective evaluations have been provided.



(a)



(b)

Fig. 4. “Google Earth Pro” images of (a) candidate points (b) the candidate points within the primary (red) threat areas identified as a result of physical effects modelling. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

- It can be said that the relative weights of the main and sub-criteria, in other words the importance levels which were evaluated for the purpose of determining the location of the assembly point, also set

out some issues for planning of facilities for organizations. The most important inferences in this context are;

- Based on the significance weight of the ‘Transportation’ criterion, the relevant establishment should be located close to the emergency

Table 7

Comparison matrices of alternatives on the basis of sub-criteria and priority vectors obtained after calculation for determining the assembly point.

Main Criterion	Sub Criteria	Weight Value/Priority Value	A	B	C	Consistency Rate
Physical Impact Area CR = 0.037 < 0.1	Worst case scenario	0.32	0.62	0.14	0.24	0.022
	Most common accident scenarios	0.23	0.68	0.12	0.22	0.033
	Primary (red) threat zone	0.17	0.62	0.24	0.14	0.022
	Secondary (orange) threat zone	0.08	0.70	0.21	0.09	0.046
	Tertiary (yellow) threat zone	0.05	0.69	0.09	0.22	0.073
	Modelling in one point	0.08	0.68	0.20	0.12	0.033
Management and Coordination CR = 0.033 < 0.1	Modelling at multiple points	0.07	0.76	0.15	0.09	0.051
	Convenience to install a crisis desk	0.19	0.26	0.63	0.11	0.048
	Close proximity to the emergency response point	0.17	0.29	0.62	0.09	0.003
	Suitability for material storage and distribution	0.17	0.31	0.58	0.11	0.003
	Internal communication	0.22	0.26	0.63	0.11	0.047
	External communication	0.25	0.67	0.24	0.09	0.009
Transportation CR = 0.021 < 0.1	Proximity to evacuation routes	0.24	0.12	0.65	0.23	0.005
	Proximity to staff intensive areas	0.10	0.14	0.72	0.14	0.000
	Access via alternative roads	0.17	0.20	0.20	0.60	0.004
	Convenience for pedestrian staff transportation	0.13	0.26	0.63	0.11	0.048
	Availability of personnel transport by vehicle	0.09	0.70	0.21	0.09	0.046
	Convenience for transportation of emergency units	0.27	0.27	0.65	0.08	0.042
Evacuation CR = 0.044 < 0.1	Ease of access outside the boundary of the establishment	0.44	0.73	0.19	0.08	0.067
	Proximity to highway transportation routes	0.20	0.73	0.19	0.08	0.067
	Proximity to the air-sea transportation route	0.09	0.12	0.23	0.65	0.005
	Proximity to emergency units	0.27	0.24	0.62	0.14	0.022
	Presence of natural water source/mains water	0.37	0.20	0.60	0.20	0.000
	Connection to sewer infrastructure	0.04	0.45	0.45	0.1	0.000
Main Requirements CR = 0.058 < 0.1	Communication infrastructure	0.34	0.28	0.65	0.07	0.058
	Proximity to the material warehouse	0.18	0.21	0.71	0.08	0.027
	Proximity to main reinforcement areas	0.07	0.26	0.67	0.07	0.024
	Area	0.09	0.43	0.43	0.14	0.000
	Ground type	0.08	0.54	0.30	0.16	0.010
	Land slope	0.06	0.50	0.25	0.25	0.000
Physical Properties CR = 0.017 < 0.1	Distance from buildings and structures	0.29	0.24	0.09	0.67	0.005
	Resistance to a secondary disaster	0.21	0.63	0.26	0.11	0.048
	Suitability to use natural barrier	0.16	0.60	0.20	0.20	0.000
	Suitability for container/tent placement	0.11	0.32	0.56	0.12	0.020
	Assembly the relevant legislation definition	0.67	0.54	0.16	0.30	0.010
	Assembly other relevant legislative needs	0.33	0.56	0.12	0.32	0.020

units and accessible from alternative routes if possible, the evacuation routes should be determined and applied alternatively.

- Based on the significance of the “physical impact area” criterion, it is necessary to identify the worst case and most common accident scenarios of the related establishment and to implement a settlement plan in which the potential threat will be kept to a minimum.
- Due to the significance weight of the “evacuation” criterion, the relevant establishment should have access to the assembly point/points, and to the safe areas outside the establishment borders from various routes.
- A systematic approach proposal based on scientific facts has been put forward for the assembly point/points determination that needs to be determined in accordance with the relevant legislation, but does not include any explanation on its selection or determination. The approach will guide all decision-makers in addition can be applied in other similar location selection problems.
- In future studies, fuzzy AHP technique can be used in order to better express the evaluation uncertainties in the binary comparison matrices for qualitative criteria during the determination of criteria weights. In the ideal assembly point selection stage, calculations can be carried out by using another MCDM method and the results can be compared with the results of this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jlp.2020.104386>.

Author contributions

Saliha ÇETİNYOKUŞ: Conceptualization, Supervision, Writing - review & editing. Onur HOŞCAN: Formal analysis, Roles/Writing - original draft.

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